Enzyme Discovery Agent

Finding enzymes for a reaction —
without ever telling the agent the answer

the model picks the strategy | the tools find the enzymes | the answer stays hidden

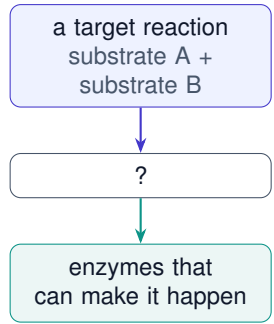
The problem

You give it a reaction — some substrates turning into a product — and it has to find enzymes that could catalyze it.

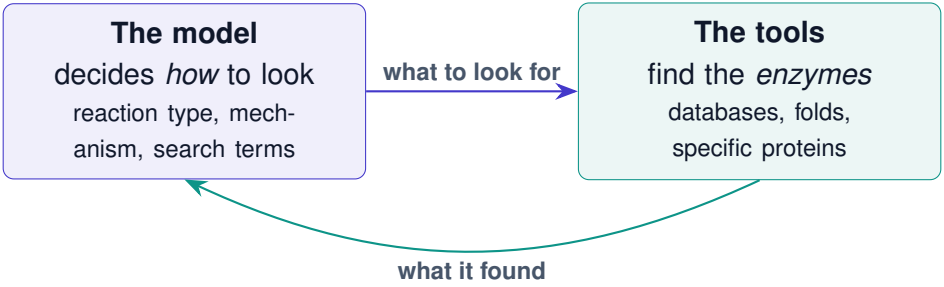
That's hard because:

- the reaction is often something **nature doesn't do**, so nothing is labelled for it
- there's nothing to look up — the answer has to be **discovered**
- and the haystack is **hundreds of millions** of proteins

The catch: hand the agent the answer's family up front, and it isn't really **discovering** anything.



The idea



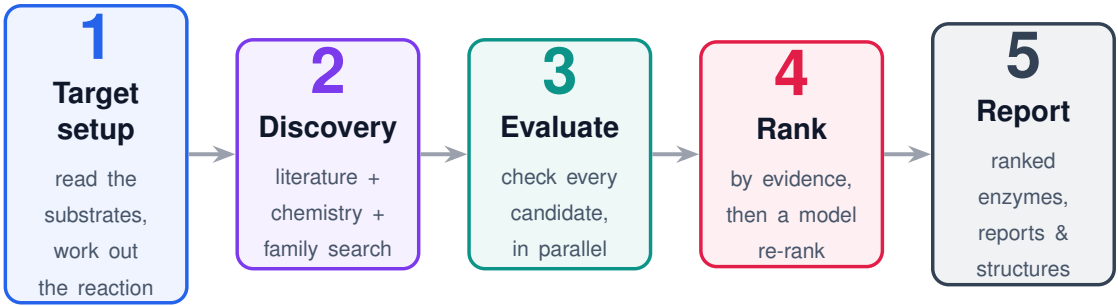
We keep the answer out of the agent:

- its enzymes, substrates, and family never appear in a prompt, a tool, or even a comment
- the real answer lives **only** in a test the agent never sees

We let the evidence do the deciding:

- the code measures the facts — direction, how well a substrate fits — and the model reasons over them
- no hand-tuned cutoffs; things are ranked against each other, not

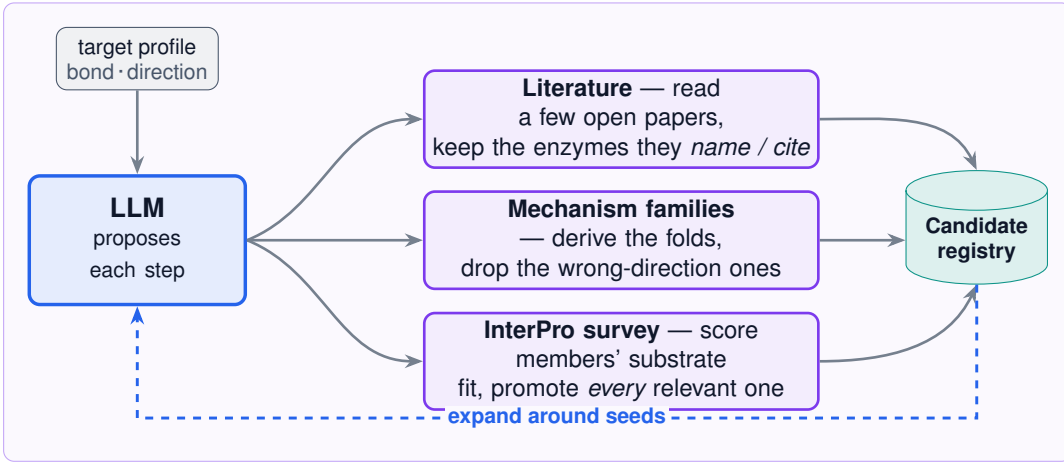
The pipeline at a glance



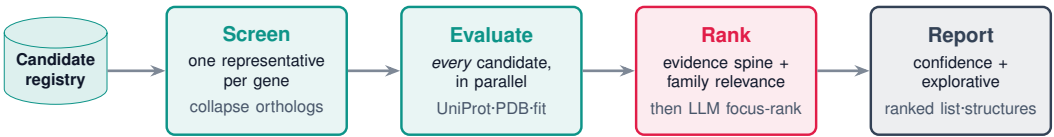
Deterministic *gates* make the important steps actually happen; the model still decides what to chase inside the loop. The next two slides open up stages 2 and 3–5.

Inside stage 2 — the discovery loop

each iteration: the model proposes → gates + tools find → the registry grows



Inside stages 3–5 — candidates to a ranked report



Evaluation uses no model calls — just database lookups — so we can afford to check *everything* and let ranking decide.

Three ways it finds candidates

From the literature

Read a few open papers and pull out the enzymes they name or cite. Often the only way to reach something that lives in a recent paper — it's how we got into the answer's **own gene cluster**.

From the chemistry

The model lists the different **folds and chemistries** that make this kind of bond, and we drop any that run the **wrong way** — a bond-breaking fold can't build one.

From the fold

For each family, look at its members, see whose natural substrate looks most **like our target**, and add the good ones. This is the main engine.

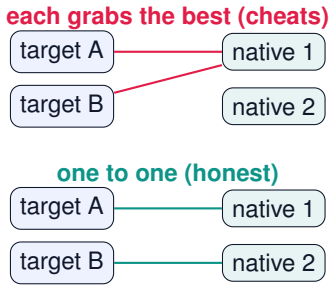
One thing we learned: judge substrate fit **member by member**, not by the family's textbook example — the two are rarely the same.

Making the substrate match honest

When the target has two substrates, we don't want it to **cheat**. If each one just grabs the enzyme's single best native, both can point at the *same* one — and a bad second partner slips by unnoticed.

So we match them **one to one**: each native can serve only one target substrate, and we score the enzyme by its **worst** pairing.

Now the target's amine is forced onto the enzyme's amine — and a tiny donor like ammonia finally scores as the poor match it is.



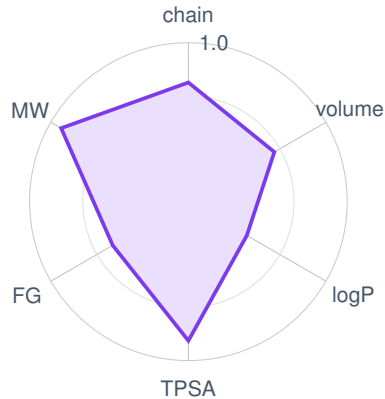
Scoring substrate fit — the method

Each candidate's native substrate is compared to the target on **six grounded physicochemical axes**, each normalised to $[0, 1]$:

- functional groups · molecular weight · main-chain length
- volume · polarity (TPSA) · hydrophobicity (logP)

For a two-substrate target we assign natives **one-to-one** and score by the **worst pair** (the bottleneck):

$$\text{fit} = \max_{\pi \in \text{1-to-1}} \min_{i} s\left(t_i, n_{\pi(i)}\right)$$



one candidate's fit across the six axes

(1.0 = identical to target)

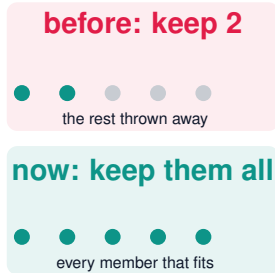
The whole point is not to miss anything

Keep every member that fits

We used to keep only a family's best two. Now every member that runs the right direction and has a real substrate to compare becomes a candidate, so a big family's long tail doesn't get thrown away.

And actually evaluate all of them

Evaluation is just database lookups, no model calls, so looking at more of them is basically free. We check every distinct gene at once — so a budget never quietly drops one that mattered.



It costs a bit more, but finding everything matters more.

How the final list is ordered

- A **fixed, reproducible core** sorts candidates by hard evidence — the bond that changes, the direction, how well the substrate fits. No magic weights.
- If the agent decided a family was **off-target**, its members sink to the bottom; members of on-target families float up.
- **Only then** does the model re-rank the short list at the top — a thin, checkable layer over an order we can already trust.

direction check → worst-substrate fit → family relevance → model re-rank

Keeping the answer out — how it's enforced

The agent must not be *told* the answer, or it isn't discovering. We enforce that as an explicit contract, not a hope:

What never appears in the agent

- answer enzymes, their substrates, or their families — in *any* prompt, tool description, or code comment
- answer-adjacent hints (naming the "right" fold or mechanism)

Where the truth lives

- only in a held-out oracle used to *grade* a run afterwards
- the agent's code is written to work for *any* target reaction

How it stays true

- the *model proposes / tools find* split: the model picks a *kind* of chemistry; the tools return the specific proteins
- a leakage `grep` over prompts + code runs on every change
- the test suite pins it (215 tests), so a regression fails loudly

Result: the benchmark answers are a *score*, never an input.

Does it work?

Our test case: find enzymes that join a six-carbon diacid and a six-carbon diamine — the building blocks of nylon-6,6. The correct answers sit in a separate test the agent never sees; we only use them to grade the run afterwards.

What comes back

- the top hits are genuine **look-alikes** — enzymes that join citrate to a diamine, chemically close to what we want
- the exact answer, **DdaG**, shows up at #4 — found through the literature, sitting in its own gene cluster
- the family list is **clean**: no bond-breaking folds, and the old habit of over-ranking a poor match is gone

And how it behaves

- a run dropped from **53 to 32 minutes** once we trimmed the reading
- almost all of that time is the **model thinking**; the tools take about 2%
- every candidate we screen now actually gets evaluated

What changed recently

Finding better enzymes

- match substrates one to one, the same way everywhere
- keep — and evaluate — every enzyme that fits
- make the family list respect reaction direction

Making it lighter

- much less literature reading, and it no longer stalls on paywalls
- evaluate candidates in parallel

Cleaning up the code

- folded nine gates into one clean pipeline
- removed duplicated scoring and direction logic
- deleted about 730 lines that weren't doing anything

all tests pass **answer still hidden**

What's still rough

- **Literature.** We still miss some of the best papers because we search with the wrong words — and a few are behind paywalls. Rewriting the queries helps, but not reliably yet.
- **Speed.** Nearly all the time is the model reading long prompts. The wins are shorter prompts and doing the final steps in parallel.
- **Ranking.** A couple of scoring formulas still don't quite agree, and we want a full run to confirm the one-to-one change helps in practice.
- **Coverage.** Make sure we really look at every member of a family, not just the ones we happen to reach.

Everything ships with the tests passing, the answer still hidden, and a note in the log.

Discover it, don’t look it up.

The model decides how to search, the tools find the enzymes, and the answer never gets to peek at itself.

honest substrate matching **keep everything that fits** **answer stays hidden**